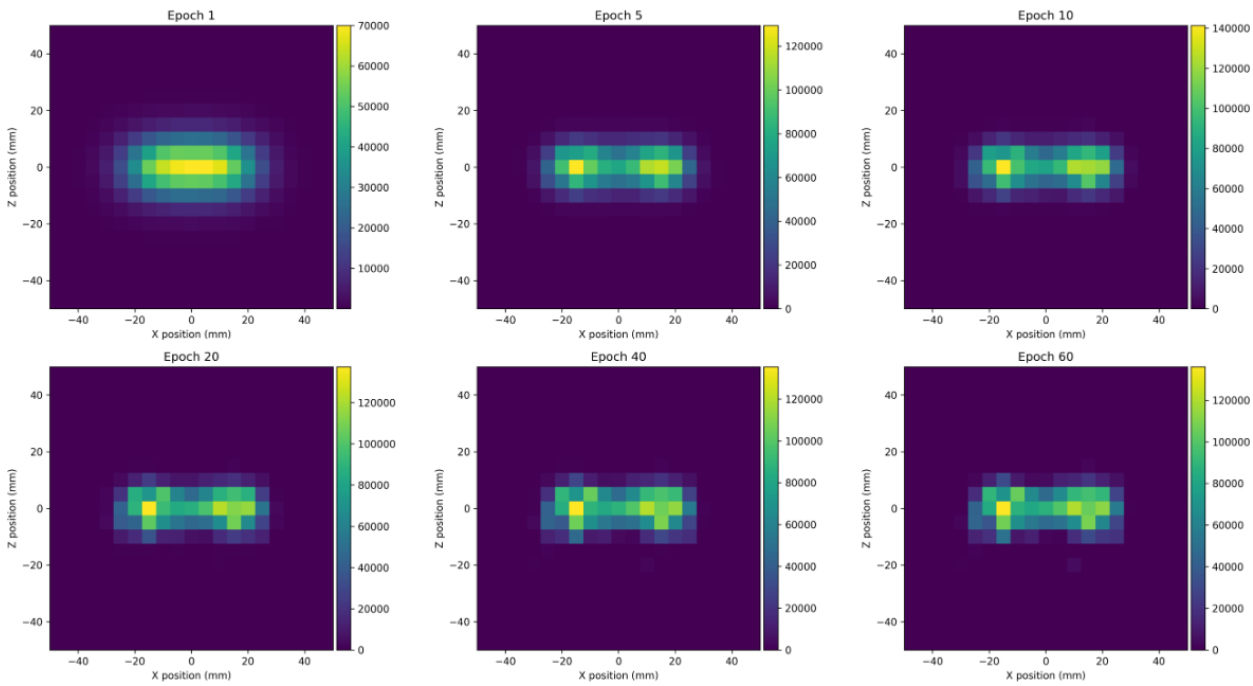
AI Assisted Therapeutic Isotope Imager with Gamma Rays

2615 TIIGR



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Executive Summary

Targeted alpha therapy (TAT) using ^{225}Ac is a promising cancer treatment, but its clinical adoption depends on the ability to image where the radiopharmaceutical accumulates in the body. Conventional SPECT imaging suffers from low sensitivity due to mechanical collimation, motivating Compton-camera imaging as an alternative that uses electronic collimation based on the physics of Compton scattering.

This project developed a simulation and reconstruction framework for a liquid-xenon Compton camera imaging the 440 keV and 218 keV gamma lines emitted in the ^{225}Ac decay chain. The framework combines GEANT4 Monte Carlo simulation, a GPU-accelerated LM-OSEM reconstruction algorithm, kernel-density-estimated system matrices, and Bayesian optimisation of event-selection parameters.

Three main outcomes were established. First, a two-point resolution study identified an intrinsic spatial resolution of approximately 30 mm for the current detector geometry at 440 keV; sources separated by 35 mm lie at the resolution threshold ($m = 0.512$) and 30 mm sources are merged ($m = 0.824$). Second, a three-part activity-concentration quality metric was developed to programmatically classify reconstructions as good or degraded, enabling automated parameter sweeps. Third, Bayesian optimisation of the scatter-energy window identified an optimal acceptance window of $[0, 122.6]$ keV, reducing the merge metric at 30 mm separation from $m = 0.824$ (uncut) to $m = 0.529$ at full convergence — a 35.8% improvement. The 30 mm separation nonetheless remains unresolved ($m > 0.5$), indicating that no two-parameter energy window can overcome the intrinsic PSF width at this source-detector distance; the optimisation correctly identifies the limits of event selection as well as its benefits.

A key finding is that attempting sub-resolution voxelisation (1 mm voxels on a PSF-limited system) produces diffuse, salt-and-pepper reconstructions due to system-matrix rank deficiency rather than insufficient statistics. Voxel size must be matched to the intrinsic point-spread function; finer resolution requires regularisation or prior information rather than brute-force event counts.

The two highest-priority recommendations for the next phase are: (1) perform a noise-free baseline study to isolate resolution contributions; and (2) conduct a symmetry and degeneracy analysis. A full set of recommendations is provided in Section 4.

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1 Introduction

1.1 Sponsor

TRIUMF, based in Vancouver, British Columbia, is Canada's national particle accelerator centre. Its work spans life sciences, radiochemistry, detector development, and scientific computing, making it well aligned with a project at the intersection of medical imaging and machine learning. The organisation's interest in this work stems from its development of the *Therapeutic Isotope Imager with Gamma Rays* (TIIGR), a Compton-camera-based system intended for imaging in targeted alpha therapy. A central challenge is that radionuclides such as actinium-225 are difficult to image directly using conventional techniques. This project is therefore directly relevant to TRIUMF's objectives, particularly in improving reconstruction quality, understanding imaging artefacts, and evaluating detector designs that may enable clinically useful imaging.

1.2 Background and Significance

1.2.1 Targeted Alpha Therapy and the Need for Imaging

Targeted alpha therapy (TAT) is a form of radiopharmaceutical therapy in which an alpha-emitting radionuclide is linked to a biological carrier that selectively targets malignant cells. The radiopharmaceutical accumulates at disease sites and emits alpha particles, which deposit large amounts of energy over very short path lengths. This combination makes TAT a promising approach for cancer treatment, enabling highly localised radiation delivery with limited exposure to surrounding healthy tissue [5, 3]. Among the radionuclides under investigation, ^{225}Ac has attracted particular interest due to its favourable half-life and decay chain, which produces multiple alpha emissions and offers considerable therapeutic potential [9, 15].

Clinical application of TAT depends on the ability to evaluate in-vivo distribution after administration: whether sufficient activity reaches the tumour, and whether uptake occurs in healthy organs at risk of toxicity. This information is essential for treatment verification, dosimetry, and individualised therapy protocols [16, 12]. Imaging therefore plays a central role in TAT, providing the basis for understanding radiopharmaceutical distribution and target uptake.

1.2.2 SPECT Imaging and Its Limitations

Single-photon emission computed tomography (SPECT) is a tomographic imaging modality that reconstructs the three-dimensional distribution of gamma-emitting radiotracers.For therapeutic imaging, SPECT supports activity quantification and absorbed-dose estimation, both important for treatment assessment and patient-specific dosimetry [16].

However, conventional SPECT has important limitations. Compared with PET, SPECT generally provides lower spatial resolution, and image quality is affected by low count statistics, attenuation, scatter, motion, and partial-volume effects [22]. Standard SPECT systems rely on mechanical collimators to obtain directional information, which rejects most incoming photons before they reach the detector and produces an inherent trade-off between spatial resolution and detection efficiency [7]. These limitations are especially significant in therapeutic imaging, where quantitative accuracy matters and available signal is already limited, motivating alternative imaging strategies for radionuclides relevant to TAT [7].

1.2.3 Compton Camera Imaging

Compton-camera imaging addresses the efficiency limitation of conventional SPECT by replacing mechanical collimation with electronic collimation based on the physics of Compton scattering. In a Compton camera, an incident gamma-ray photon undergoes a Compton interaction in a first detector (the scatter detector) and is then absorbed in a second detector. From the deposited energies and interaction positions, the photon's origin is constrained to a cone of possible source locations (the Compton cone). Combining information from many such cones reconstructs the three-dimensional source distribution.

The scattering angle follows the Compton kinematic relation

$$\cos\theta_c = 1 - \frac{m_e c^2\, E_C}{E_\gamma (E_\gamma - E_C)},$$

where E_γ is the incident photon energy, E_C is the energy deposited in the Compton scatter, m_e is the electron mass, and c is the speed of light. This relation links measured detector signals to the geometric constraint used in reconstruction. Unlike SPECT's mechanical collimation, Compton cameras accept a far larger fraction of incident photons, providing higher sensitivity at the cost of a more complex reconstruction problem.

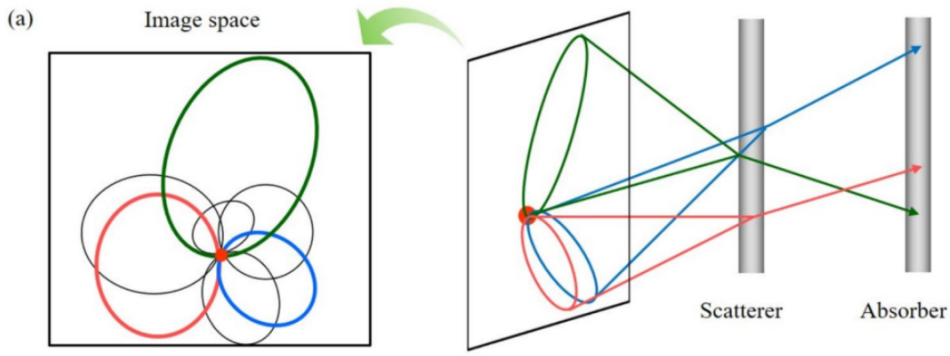

Figure 1: Compton camera imaging principle. Reproduced from [13].

1.3 Project Overview and Scope

This project investigates image reconstruction for radionuclide imaging in the context of TAT, with emphasis on a Compton-camera framework for the TIIGR detector. The overall aim is to improve estimation of radioactive source distributions through a computational pipeline combining GEANT4 Monte Carlo simulation, detector-response modelling, and iterative image reconstruction. A central focus is improving computational efficiency without compromising reconstruction quality, including GPU-accelerated reconstruction and Bayesian optimisation of event-selection parameters. Detailed GEANT4 configuration is provided in Appendix B. The physical justification for simulation event counts is given in Appendix E.

Three aspects are deliberately outside the scope of this phase. First, realistic patient anatomy (the water-box phantom) was not activated: the priority was to establish the reconstruction framework and characterise baseline performance in a clean, attenuation-free geometry before introducing tissue scatter that would conflate algorithmic and physical contributions to image quality. This is a deliberate sequencing decision; the framework already includes the water-box phantom and activating it is a defined next step (Recommendation 5). Second, the full ^{225}Ac decay chain was replaced by monoenergetic photon generation at 440 keV and 218 keV, a design choice to enable clean benchmarking at each gamma line in isolation. Mixing multi-energy coincidences and daughter-nuclide contributions at this stage would have made it impossible to attribute reconstruction artefacts to specific physical causes. Third, experimental validation against a physical detector prototype was not performed because no physical prototype currently exists; the simulation framework and quality metrics developed here may serve as a starting point for future

acceptance testing once a prototype is constructed.

1.4 Project Goals

The project had four specific objectives:

1. **Build a validated simulation pipeline.** Develop an end-to-end GEANT4-to-reconstruction workflow for a two-layer liquid-xenon Compton camera, with reproducible configuration management and ROOT/HDF5 data interchange between stages.
2. **Characterise spatial resolution.** Quantify the detector's intrinsic spatial resolution at 440 keV using two-point source simulations, and compare against a first-principles estimate from Compton angular uncertainty.
3. **Develop automated quality metrics.** Define quantitative metrics for reconstruction quality that can be evaluated programmatically, enabling parameter sweeps and optimisation without manual image inspection.
4. **Optimise event selection.** Apply Bayesian optimisation to the scatter-energy acceptance window and assess whether principled selection yields meaningful improvement over default cuts.

2 Discussion

2.1 Theory and Fundamentals

2.1.1 System Matrix and LM-OSEM Reconstruction

The system matrix A describes how a source distribution maps to detector measurements. Dividing object space into voxels with unknown source activity $\boldsymbol{\lambda}$, each matrix element $A_j(y_n)$ gives the probability that event y_n originated from voxel j. The system matrix is therefore the forward model, incorporating detector geometry, interaction physics, energy response, and event-selection criteria.

Within the central field of view, cylindrical symmetry makes the detector response approximately position-independent, so $A_j(E_C, \theta) \approx A(E_C, \theta)$. The system matrix is built from a

volume source simulation that uniformly samples positions within the field of view, rather than point-by-point calibration simulations. Separate system matrices are constructed for 440 keV and 218 keV.

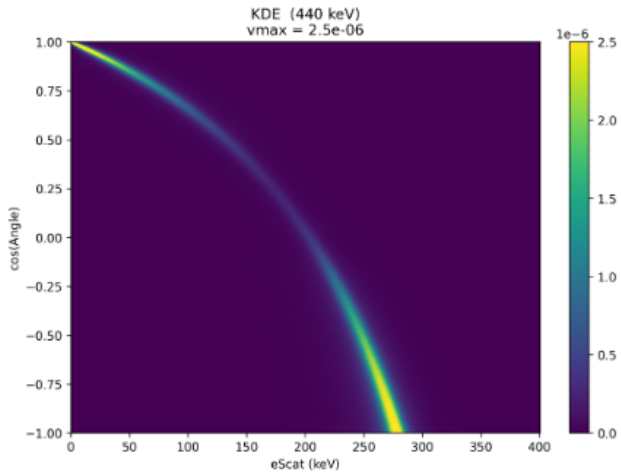

Figure 2: 440 keV system-matrix density $P(E_C, \cos\theta)$ estimated by two-dimensional kernel density estimation (Epanechnikov kernel) from a volume-source simulation with photons uniformly distributed within a 30 mm-radius sphere. Horizontal axis: scatter energy E_C deposited in the inner detector layer (keV). Vertical axis: cosine of the Compton scattering angle θ_c derived from the kinematic relation $\cos\theta_c = 1 - m_e c^2 E_C/[E_\gamma(E_\gamma - E_C)]$. The ridge at low E_C / high $\cos\theta_c$ reflects near-forward scattering; the high-E_C tail at $\cos\theta_c \approx -1$ corresponds to $180°$ backscattering. In LM-OSEM reconstruction this density serves as the forward model $A(E_C, \theta)$: for each measured event the system-matrix value approximates the probability of detecting that $(E_C, \cos\theta_c)$ pair from any source position within the field of view, under the position-invariant approximation.

Reconstruction uses list-mode ordered-subsets expectation maximisation (LM-OSEM), an iterative algorithm for emission tomography. In list-mode, data are processed event-by-event rather than grouped into projection bins, which suits Compton-camera imaging where each photon carries its own measured energy and geometry. The update rule is

$$\lambda_j^{k+1} = \frac{\lambda_j^k}{\xi_j} \sum_{n \in s} \frac{A_j(y_n)}{\sum_l A_l(y_n)\lambda_l^k},$$

where λ_j^k is the estimated activity in voxel j at iteration k, s is the current subset, and ξ_j is a sensitivity normalisation. Ordered subsets accelerate convergence by updating the image using smaller groups of events within each iteration. The voxel grid is initialised uniformly and run for 60 epochs, with the full 3D estimate saved after each epoch.

2.1.2 Kernel Density Estimation

Kernel density estimation (KDE) is a non-parametric method for approximating an unknown probability density from finite samples. Rather than binning data, KDE places a smooth kernel at each sample point and sums the contributions. For samples $x_1, \ldots, x_N \in R^d$,

$$\hat{f}(x) = \frac{1}{Nh^d} \sum_{i=1}^{N} K\left(\frac{x - x_i}{h}\right),$$

where K is the kernel and h is the bandwidth. Smaller h preserves local detail at the cost of noise; larger h smooths at the cost of resolution. KDE is applied in two dimensions over $(E_C, \cos\theta)$ using an Epanechnikov kernel, implemented via scikit-learn's `KernelDensity`. The fitted density is evaluated over the $(E_C, \cos\theta)$ domain to form a smooth approximation of the system matrix, which is more stable than direct binning when Monte Carlo statistics are limited.

2.1.3 Spatial Resolution: FWHM

Spatial resolution is quantified by the full width at half maximum (FWHM) of the reconstructed activity profile, obtained by projecting through the image centre and measuring the width at half the peak value. Smaller FWHM indicates better resolution. FWHM is computed independently along each Cartesian axis and tracked as a function of epoch to monitor convergence.

2.2 Simulation Pipeline

The workflow consists of five stages, each producing intermediate data files that feed the next. System-matrix construction (Stage 2) is a one-time pre-computation that is independent of the source being imaged; it need only be repeated if the detector configuration or gamma energy changes.

Stage 1 — Monte Carlo simulation. GEANT4 generates photon histories from a source configuration defined in a macro file. Two classes of simulation are run: a spatially uniform volume source used for system-matrix construction (Stage 2), and the source geometry of interest used for reconstruction (Stage 3). Each run produces a ROOT file containing per-event interaction data and run metadata.

Stage 2 — System-matrix construction. Volume-source ROOT files are processed through the event-selection and detector-smearing (Appendix C) pipeline according to a YAML configuration, and the conditional distribution $P(E_C, \cos\theta)$ is estimated by KDE. The result is stored as a 2D array in HDF5 with full provenance metadata.

Stage 3 — Data conversion. ROOT files from the source of interest are processed through the same event-selection pipeline, producing a reconstruction-ready HDF5 data sample. Each surviving event is stored with smeared scatter and absorption positions and combined energies. The detection efficiency $\varepsilon = N_{\text{final}}/N_{\text{simulated}}$ and the full processing configuration are stored as file attributes.

Stage 4 — Image reconstruction. The LM-OSEM algorithm reads the HDF5 data sample and the pre-computed system matrix and executes on GPU. One HDF5 file is written per iteration, storing the voxel activity map and enabling resumption from any epoch.

Stage 5 — Analysis and visualisation. Post-processing scripts produce 2D centre slices (YZ, XZ, XY), 1D activity profiles, FWHM convergence curves, and two-point source merge-detection metrics. Table 1 summarises formats and storage locations.

Stage	**Format**	**Location**	**Key contents**
GEANT4 output	ROOT	`DATA_FILES/{E}/`	Per-interaction data, run metadata
System matrix	HDF5	`SYSTEM_MATRIXES/`	$P(E_C, \cos\theta)$ array, provenance
Data sample	HDF5	`DATA_SAMPLES/`	Per-event (r_C, r_A, E_C, E_A), sensitivity
Reconstruction	HDF5	`OUTPUT/`	3D activity array, config, epoch

Table 1: File formats and storage locations at each pipeline stage.

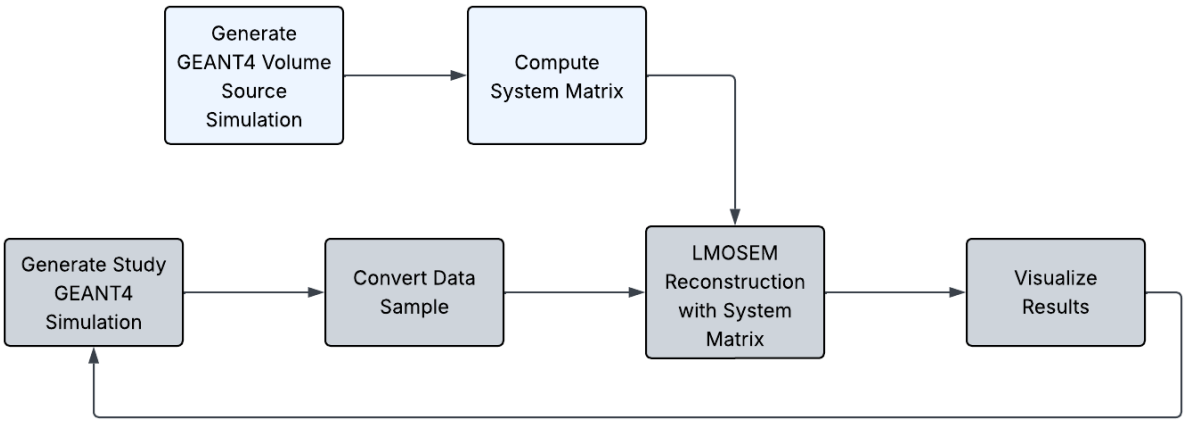

Figure 3: End-to-end simulation and reconstruction pipeline, showing the five processing stages from GEANT4 simulation to analysis output. Arrows indicate data flow between stages; file formats produced at each stage are summarised in Table 1. The system matrix (Stage 2) is computed once from the volume-source simulation and reused across all subsequent reconstructions.

2.2.1 Volume Source and System-Matrix Construction

The system matrix is constructed from a volume source simulation in which gamma photons are emitted isotropically from positions sampled uniformly within a sphere of radius 30 mm centred at the origin. Uniform volume sampling uses the cube-root transform: for $u \sim U[0, 1]$, the radial position $r = R \cdot u^{1/3}$ gives positions uniformly distributed through the sphere volume, correcting for the $r^2\, dr$ volume element that would otherwise over-represent the centre.

The volume source is used exclusively for system-matrix construction and does not represent a specific patient scenario. By densely covering the field of view, it provides a representative sample of the detector response at all source positions, supporting the position-invariant approximation. Multiple independent simulation runs with different random seeds accumulate sufficient statistics for accurate density estimation.

2.2.2 Event Selection

Raw GEANT4 output contains the full photon history for every simulated event. A sequence of topological and geometric cuts selects events useful for Compton-camera reconstruction:

1. **No air interactions.** All interactions must occur within a detector volume ($\forall\, i$: $\texttt{volumeID}_i \neq 0$).

2. **Interaction multiplicity.** At least two interactions are required ($n \geq 2$), with the first being a Compton scatter and the final a photoelectric absorption. Additional interaction vertices are permitted provided they lie within the absorption cluster (see cut 5), consistent with standard practice in modern liquid-xenon Compton cameras [11, 23]. Enforcing strict $n = 2$ would reject characteristic Xe K-shell fluorescence photons that are an unavoidable product of photoelectric absorption, reducing efficiency with no physics benefit.

3. **Scatter in inner detector.** The Compton scatter must occur in the inner detector layer ($\texttt{volumeID}[0] = 1$), ensuring a well-defined scattering geometry for cone reconstruction.

4. **Scatter-to-absorption distance.** The distance between the Compton scatter vertex and the photoelectric absorption vertex must exceed 100 mm, rejecting events in which the two interaction sites are insufficiently separated to reconstruct a reliable cone direction.

5. **Absorption cluster size.** The photoelectric absorption must be spatially contained within a 3 mm radius, selecting clean single-site depositions and rejecting diffuse clusters.

2.3 Spatial Resolution Characterisation

2.3.1 Two-Point Source Resolution Study

Pairs of 440 keV point sources were simulated symmetrically along the x-axis at separations d from 1 mm to 50 mm, with each source placed at $(\pm d/2, 0, 0)$. Unless otherwise noted, each source was simulated with 10^7 primary photons. Since GEANT4 supports only one source position per run, each source was simulated independently and the resulting ROOT files were merged at the data conversion step.

Reconstructions with 5 mm voxels showed the two sources clearly distinguishable at 50 mm and 40 mm, progressively converging as separation decreased, and beginning to merge at approximately 30 mm — indicating the system's resolution limit under this detector configuration.

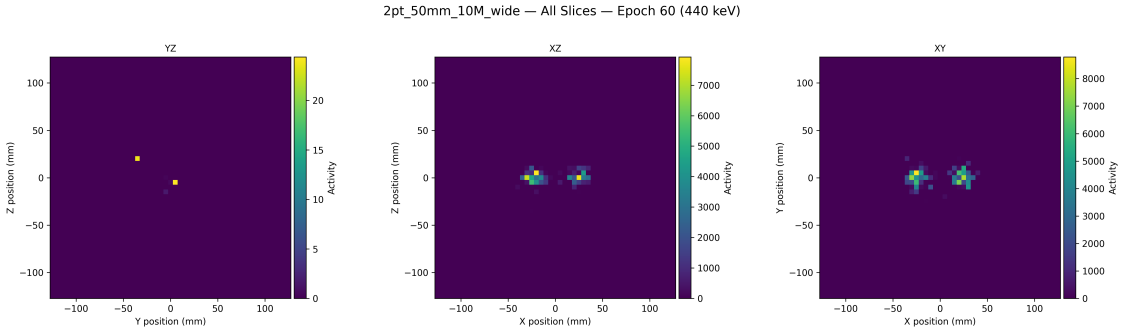

Figure 4: Reconstruction of two 440 keV point sources separated by 50 mm (5 mm voxels, epoch 60). Sources are clearly resolved; merge metric $m = 0.029$.

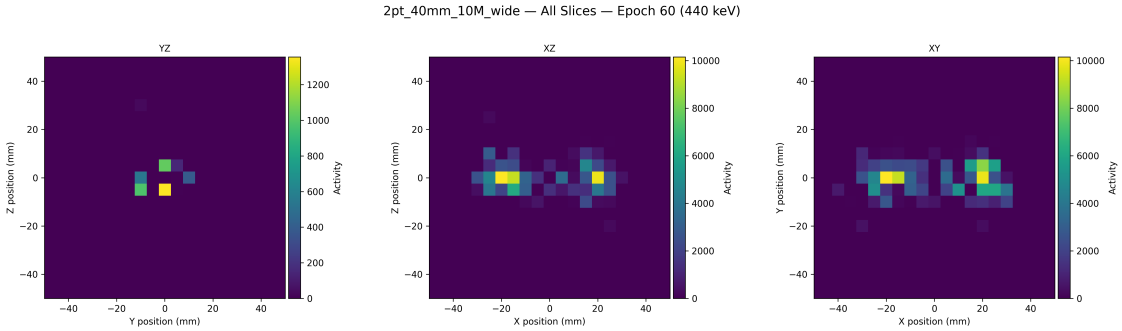

Figure 5: Reconstruction at 40 mm separation (5 mm voxels, epoch 60). Sources remain distinguishable; merge metric $m = 0.211$, below the Rayleigh-criterion threshold of 0.5.

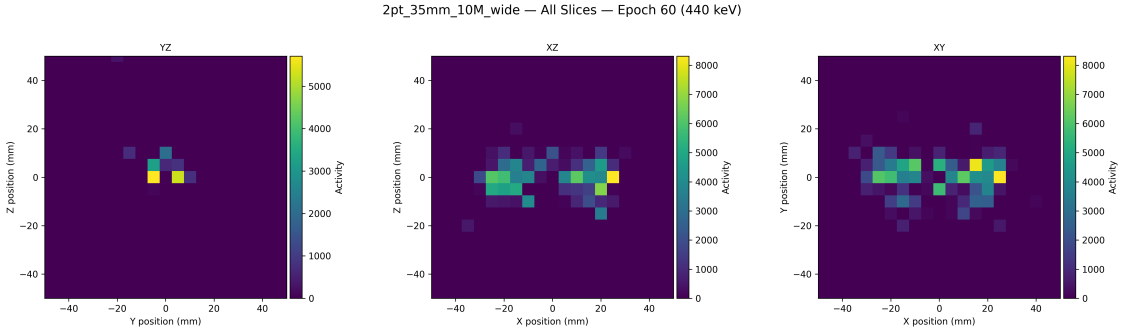

Figure 6: Reconstruction at 35 mm separation (5 mm voxels, epoch 60). Peaks visibly narrow toward each other; merge metric $m = 0.512$, at the Rayleigh-criterion boundary.

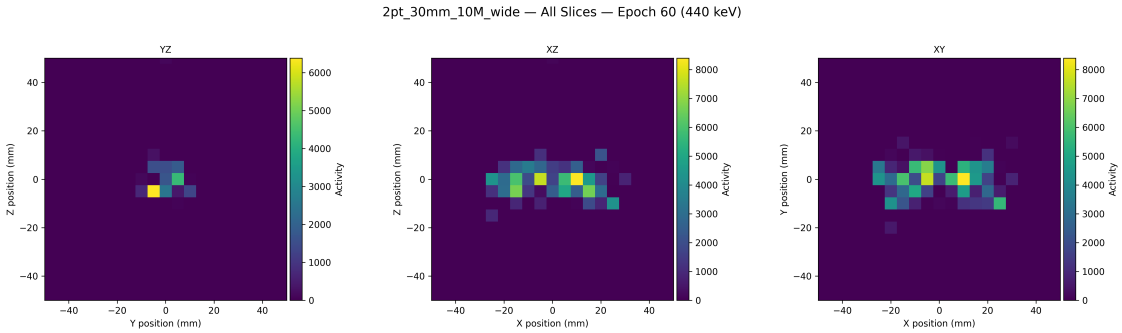

Figure 7: Reconstruction at 30 mm separation (5 mm voxels, epoch 60). The two peaks merge into a single lobe; merge metric $m = 0.824$, confirming the system has reached its resolution limit.

To test whether merging was a statistical artefact, the event count was increased to 2.5×10^8 per source at 30 mm separation. Counterintuitively, merging became *more* pronounced:

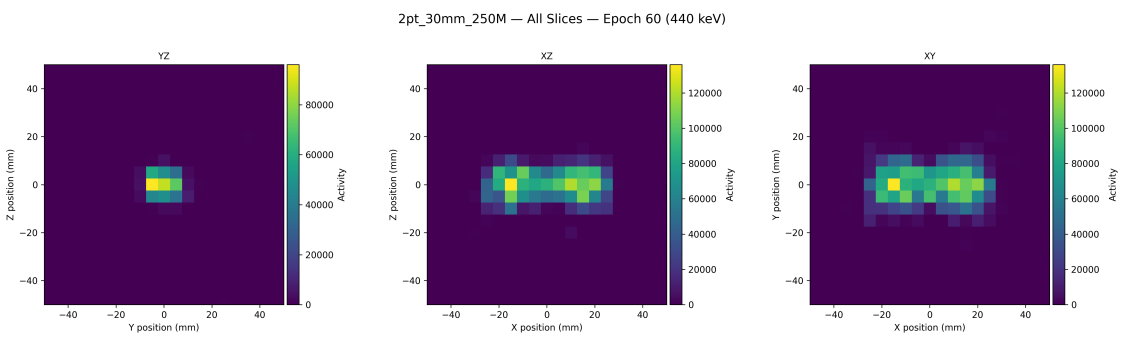

Figure 8: Reconstruction at 30 mm separation with 2.5×10^8 events per source (5 mm voxels, epoch 60). Merging is more pronounced than at lower statistics, confirming the 30 mm limit is a fundamental system property rather than a data-starvation artefact.

This behaviour follows directly from how maximum-likelihood EM algorithms respond to increasing statistics. As the event count grows, the algorithm converges to the true ML image, which at sub-resolution separations is precisely the merged single-lobe solution. Observing stronger merging at higher statistics therefore confirms that the 30 mm limit is a fundamental system property, not a data-starvation artefact.

The intrinsic angular resolution of the Compton camera provides a foundation for this limit. Differentiating the Compton relation gives the propagated angular uncertainty due to finite energy resolution δE_C:

$$\delta\theta_c = \frac{m_e c^2}{\sin\theta_c \, (E_\gamma - E_C)^2} \, \delta E_C.$$

The corresponding spatial blur at a source distance r from the inner detector is $\delta r \approx r\,\delta\theta_c$. For the detector geometry used here ($r = 300$ mm), even modest energy resolution broadens the effective point-spread function to multi-centimetre scale. Additional geometric contributions arise from the finite depth of interaction within the 0.60 X_0 inner layer and the angular binning of the system matrix. Together, these effects set the intrinsic resolution, consistent with the empirically observed $\sim$30 mm limit.

Comparison with published Compton-camera systems. The 30 mm result can be contextualised against the published literature on Compton cameras for ^{225}Ac-chain imaging. Caravaca et al. [4] demonstrated a CZT-based Compton camera achieving (1.58 ± 0.03) mm FWHM for ^{213}Bi at 440 keV, with an angular uncertainty of $(5.10 \pm 0.10)^\circ$. The critical difference is source-detector distance: the Caravaca system operates at $r \approx 15.5$ mm (a preclinical mouse-imaging geometry), whereas the TIIGR inner detector radius is $r = 300$ mm. Since spatial blur scales as $\delta r \approx r\,\delta\theta_c$, projecting the Caravaca angular uncertainty to the TIIGR geometry gives $\delta r \approx 300\,\mathrm{mm} \times \sin(5.1^\circ) \approx 27$ mm, in close agreement with the observed $\sim$30 mm limit. The TIIGR detector is therefore not inferior in angular resolution to the Caravaca system; the resolution difference reflects the substantially larger field of view required for human-scale imaging. Achieving sub-centimetre resolution at $r = 300$ mm would require reducing $\delta\theta_c$ by a factor of $\sim$30 relative to the Caravaca benchmark.

2.3.2 Effect of Voxel Size

Reconstructions were repeated at 30 mm separation using 1 mm voxels over the same volume. The output was dominated entirely by noise, with no coherent structure visible (Figure 9).

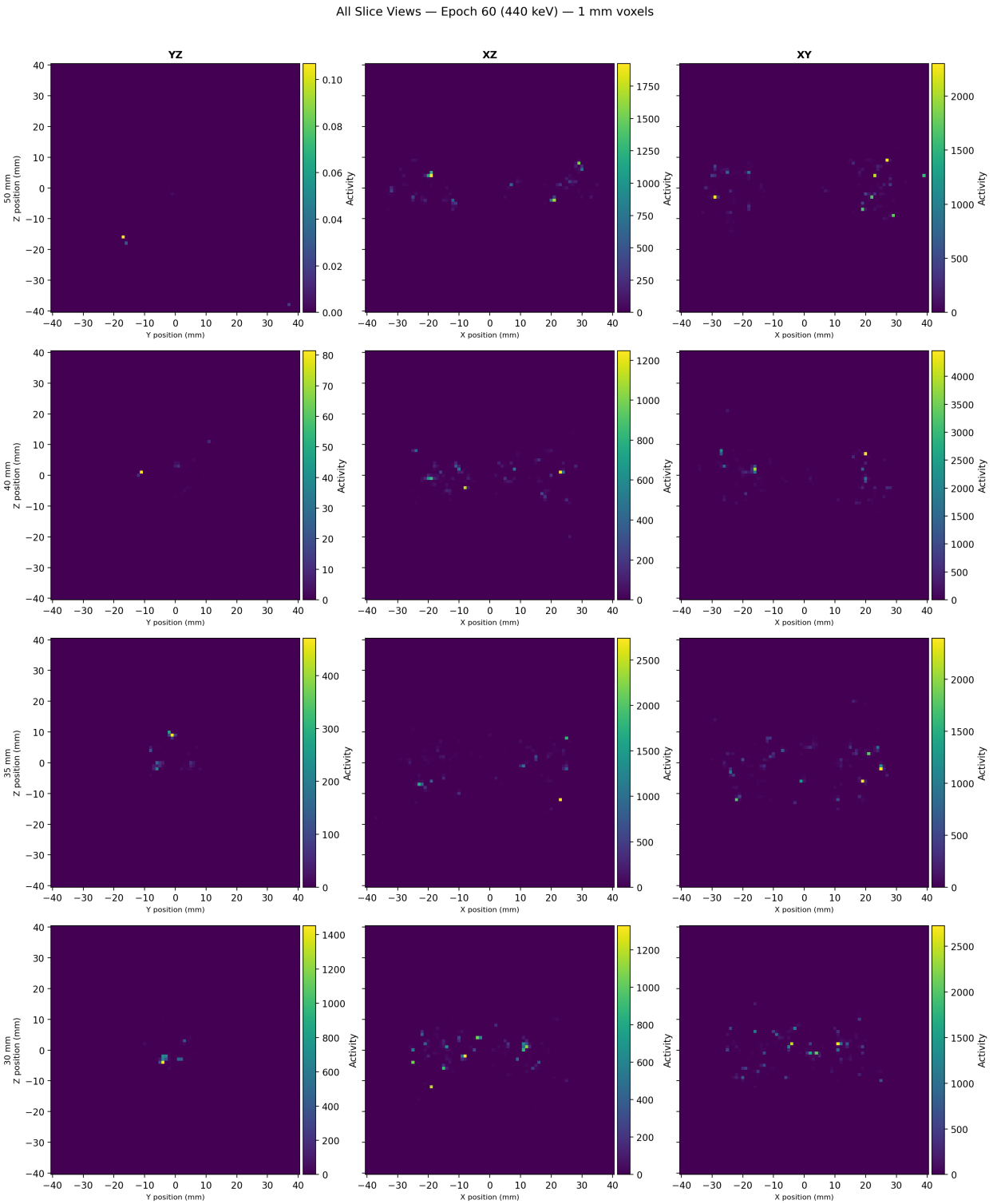

Figure 9: 1 mm voxel reconstructions at source separations of 30, 35, 40, and 50 mm (epoch 60, 440 keV). The salt-and-pepper pattern is visible at every separation, confirming that noise dominates regardless of source spacing when voxels fall below the system PSF. This failure mode arises from system-matrix rank deficiency rather than insufficient statistics (see text).

This outcome follows from two compounding effects.

Statistical undersampling. At $\varepsilon \approx 1.67\%$ detection efficiency, two sources each simulated with 10^7 primary photons yield approximately 3.34×10^5 detected events. Over a $31 \times 31 \times 31$ mm^3 volume at 1 mm voxels, this gives 29,791 voxels with mean count $\bar{n} \approx 11.2$ and SNR ≈ 3.3. At 5 mm voxels over the same volume, ~ 216 voxels receive $\sim 1{,}546$ events each (SNR ≈ 39), providing stable convergence.

Sub-resolution parameterisation. The more fundamental issue is that the 31 mm volume is commensurate with the system PSF (~ 30 mm FWHM). Virtually every pair of 1 mm voxels lies within one resolution element of another, so the system-matrix rows $A_j(E_C, \theta)$ are nearly identical across the grid. The matrix is effectively rank-deficient: the reconstruction has 29,791 unknowns but only $\mathcal{O}(1)$ independent pieces of spatial information. LM-OSEM distributes activity arbitrarily among degenerate voxels at each iteration, amplifying Poisson noise rather than converging. Appropriate voxel size is set by the system PSF, not geometric preference: sub-resolution voxelisation cannot recover structure the physical system cannot resolve, regardless of event count.

2.4 Evaluation Metrics

2.4.1 Merge Metric

To quantify whether two closely spaced sources are resolved or merged, a continuous valley-to-peak metric is computed on the x-axis marginal profile of the 3D activity volume. The x-axis is the projection axis since both sources lie along it; summing over transverse dimensions improves profile SNR by $\sqrt{n_y n_z}$ relative to a single row.

The marginal profile is

$$p(x_i) = \sum_{j,k} \hat{\lambda}(x_i, y_j, z_k),$$

convolved with a Gaussian kernel ($\sigma = 5$ mm) to suppress sub-voxel noise, then normalised to $[0, 1]$. Peaks are identified with minimum height 0.15 and minimum separation 8 mm. For two peaks at x_L, x_R, the merge metric is

$$m = \frac{p_{\text{valley}}}{\min\{p(x_L), p(x_R)\}}, \quad m \in [0, 1],$$

where $p_{\text{valley}} = \min_{x_L \le x \le x_R} p(x)$. Normalising by the smaller peak ensures $m \le 1$ for

asymmetric reconstructions. Sources are declared resolved when $m < 0.5$, analogous to the Rayleigh criterion. The continuous metric over $[0, 1]$ is directly suitable as a Bayesian optimisation objective.

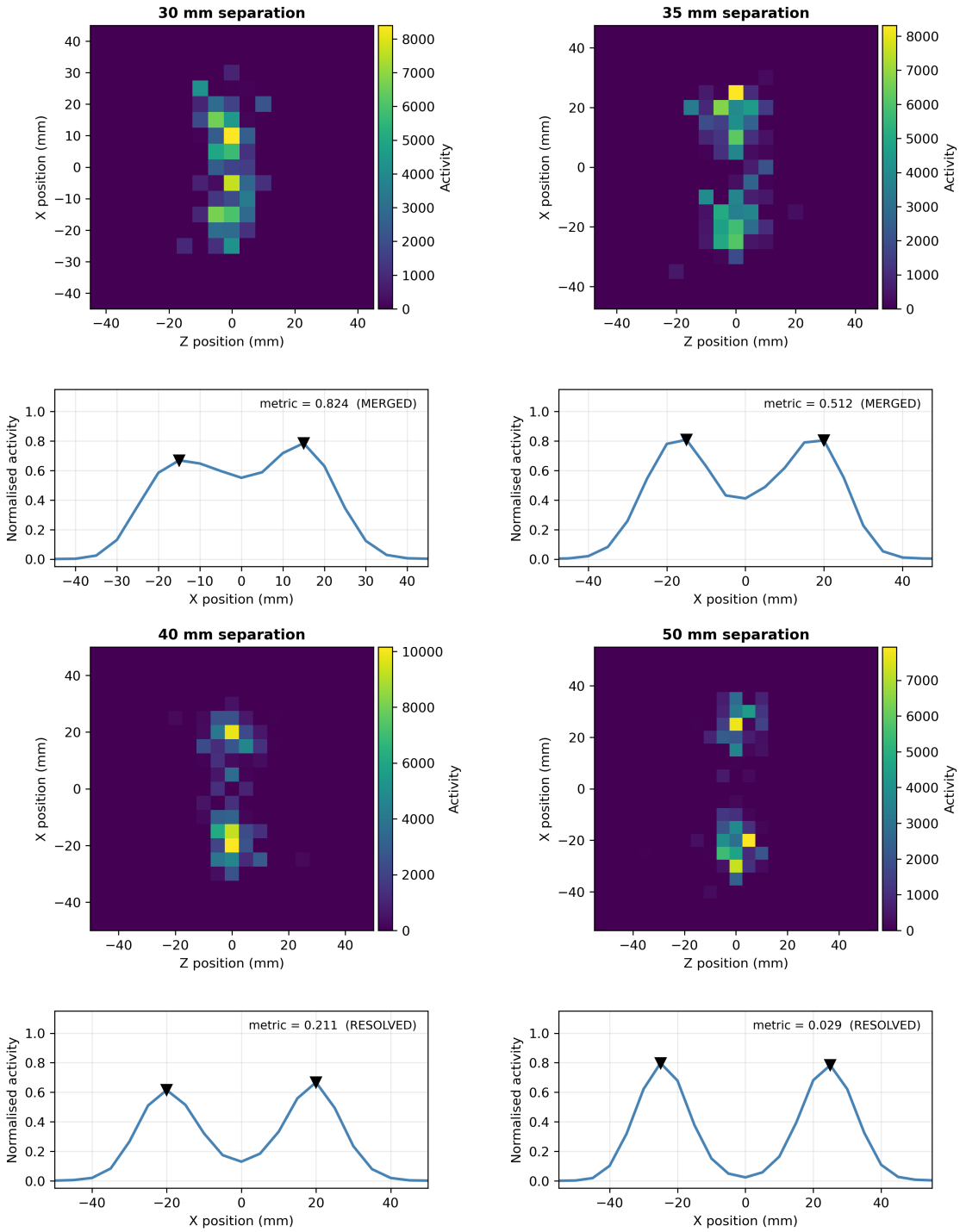

Figure 10: Merge metric evaluated across two-point source separations. Upper panels: XZ centre slice of reconstructed activity at epoch 60 (5 mm voxels, 440 keV, KDE system matrix). Lower panels: normalised x-axis marginal profile with triangles marking detected peaks. At 30 mm ($m = 0.824$) and 35 mm ($m = 0.512$) the sources are merged under the Rayleigh criterion ($m \geq 0.5$); at 40 mm ($m = 0.211$) and 50 mm ($m = 0.029$) they are resolved.

2.4.2 Activity-Concentration Quality Metric

At sub-resolution voxel sizes or source separations near the resolution limit, LM-OSEM can fail to localise activity and instead distribute it diffusely, producing salt-and-pepper artefacts [20, 18]. Visual identification of this failure mode is easy, but programmatic classification is needed for parameter sweeps. Since source positions are known in simulation, ground truth can be exploited to define three complementary quality metrics. Full threshold calibration is given in Appendix F.

Integration region. For each source position $\mathbf{s}_p$, define a cubic region $\mathcal{C}_p$ of half-width $h = 10\,\text{mm}$. The full integration region is $\mathcal{C} = \bigcup_p \mathcal{C}_p$, with background $\mathcal{B} = \mathcal{V} \setminus \mathcal{C}$. The half-width $h = 10\,\text{mm}$ is chosen empirically: well-converged 5 mm-voxel reconstructions consistently concentrate $\sim 88\%$ of total activity within this region.

Metric 1: Ratio.

$$\rho = \frac{\Lambda_{\text{local}}}{\Lambda_{\text{total}}} = \frac{\sum_{(i,j,k)\in\mathcal{C}} \hat{\lambda}_{ijk}}{\sum_{\text{all}} \hat{\lambda}_{ijk}}, \quad \rho \in [0, 1].$$

Measures what fraction of total activity falls within the integration region. Threshold: $\rho_{\text{thresh}} = 0.5$.

Metric 2: Contrast.

$$C = \frac{\bar{\lambda}_{\text{local}}}{\bar{\lambda}_{\text{bg}}} = \frac{\Lambda_{\text{local}}/|\mathcal{C}|}{(\Lambda_{\text{total}} - \Lambda_{\text{local}})/|\mathcal{B}|}.$$

Measures how much brighter source voxels are than average background voxels [2]. Empirically, 5 mm reconstructions yield $C \approx 2000$–$4000\times$; 1 mm reconstructions yield $C \approx 80$–$120\times$. Threshold: $C_{\text{thresh}} = 200\times$.

Metric 3: Hot-background fraction.

$$\phi = \frac{1}{|\mathcal{B}|} \sum_{(i,j,k)\in\mathcal{B}} \mathbf{1}\left[\hat{\lambda}_{ijk} > \alpha \frac{\Lambda_{\text{total}}}{|\mathcal{C}|}\right].$$

Counts scattered bright voxels that contrast alone misses [20]. Threshold: $\phi_{\text{thresh}} = 0.20\%$ with $\alpha = 0.05$.

Combined flag. A reconstruction is declared GOOD only if all three criteria pass:

$$Q_{\text{combined}} = \mathbf{1}[\rho \geq 0.5] \cdot \mathbf{1}[C \geq C_{\text{thresh}}] \cdot \mathbf{1}[\phi \leq \phi_{\text{thresh}}].$$

Validation against nine configurations (two voxel sizes, multiple separations, two epoch

counts) correctly classified every case. The 1 mm-voxel reconstructions, which would pass ρ alone, are correctly identified as BAD by both contrast and hot-background fraction. An adaptive variant of the ratio threshold for high-coverage edge cases, in which the mask occupies a substantial fraction of the reconstruction volume, is described in Appendix F.

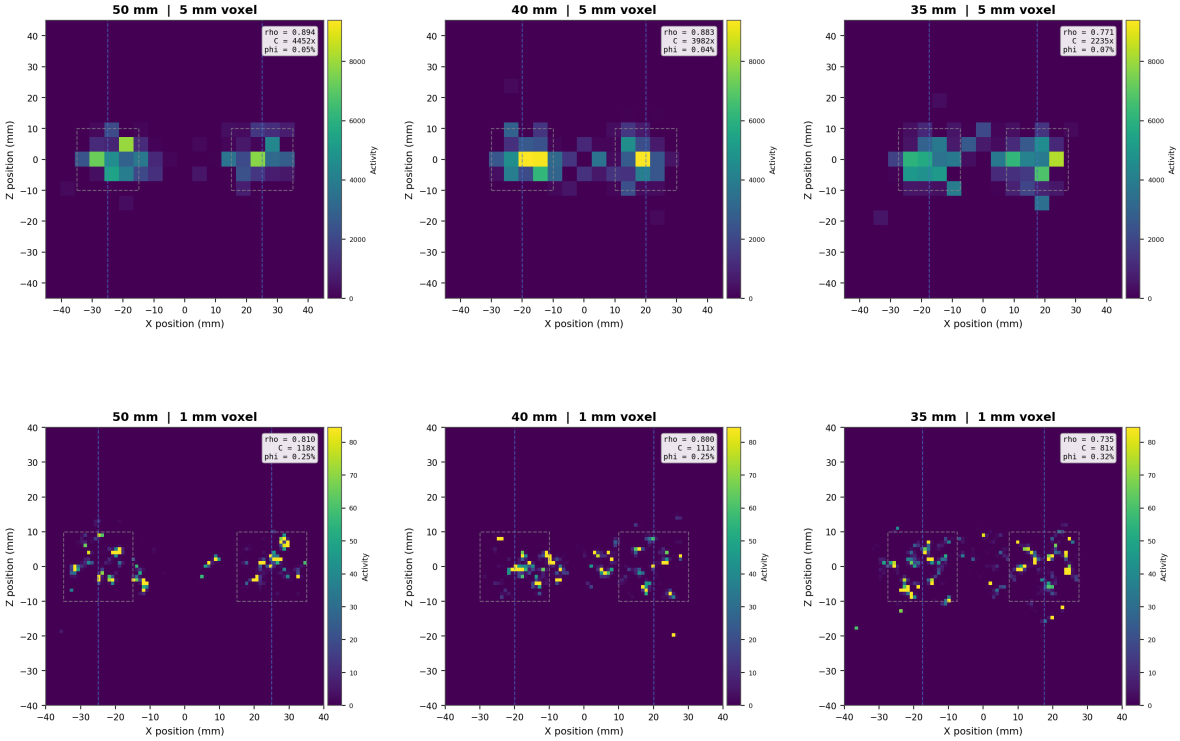

Figure 11: Central XZ slices of LM-OSEM reconstructions at separations of 35–50 mm, for 5 mm voxels (top row) and 1 mm voxels (bottom row) at epoch 60. The 5 mm reconstructions are classified GOOD by all three quality metrics. The 1 mm reconstructions exhibit salt-and-pepper noise and are classified BAD by the contrast and hot-background fraction metrics, even though their ratio metric ρ alone would pass.

Limitations. The metric requires known ground-truth source positions and is therefore applicable only in simulation. The thresholds ($C_{\text{thresh}} = 200\times$, $\phi_{\text{thresh}} = 0.20\%$) were calibrated on nine reconstruction configurations for this specific detector geometry and event counts, and may require retuning for substantially different configurations; the small validation set limits confidence in threshold generalisability. The metric cannot distinguish a merged reconstruction from a scattered one, since both reduce ρ; however, merged reconstructions are correctly identified as BAD by the combined metric regardless.

2.5 Bayesian Optimisation

2.5.1 Energy-Cut Optimisation

Not all detected Compton events carry equal spatial information. Events with very small scatter energy E_C correspond to near-forward scattering and poorly constrain source position, while events with large E_C are dominated by angular uncertainty amplified through the steep slope of the Compton kinematic relation. Restricting reconstruction to an energy window $[E_{C,\min}, E_{C,\max}]$ therefore improves effective resolution at the cost of reducing event count.

The objective of this optimization problem is the merge metric m as a function of window edges. This expensive to evaluate, has no closed-form gradient, and cannot be assumed unimodal. Thus grid search is computationally intractable. Bayesian optimisation is therefore applied as a sample-efficient strategy for black-box hyperparameter search.

The Bayesian Optimization framework models $m(E_{C,\min}, E_{C,\max})$ as a Gaussian process with a Matérn-5/2 covariance kernel, selects the next trial by maximising the Expected Improvement acquisition function, and re-estimates kernel hyperparameters after each trial by maximising the marginal likelihood. The merge metric is treated as deterministic, since Gaussian smoothing suppresses the small trial-to-trial variation from LM-OSEM batch ordering. Full mathematical detail is given in Appendix D.

A hard minimum event count of $N_{\min} = 500$ acts as an implicit barrier: narrower windows receive the maximum penalty $m = 1.0$ without performing a reconstruction. The feasible parameter space is constrained by $E_{C,\min} \in [0, 220]$ keV, $E_{C,\max} \in [80, 400]$ keV, and $E_{C,\max} \geq E_{C,\min} + 40$ keV. The first six trials use a quasi-random Sobol sequence; subsequent trials are selected by the acquisition function. Each trial runs LM-OSEM for 20 epochs and evaluates the merge metric on the final epoch. Experiment state is serialised to disk after every trial for crash recovery. The framework uses the `ax-platform` library with the BoTorch back-end [1].

Results. Thirty trials were completed, spanning $E_{C,\min} \in [0, 220]$ keV and $E_{C,\max} \in [80, 400]$ keV on the 30 mm separation dataset. The optimal window identified by the GP surrogate was $[0, 122.6]$ keV, corresponding to Compton scattering angles $\theta_c \in [0°, 57°]$ (forward-to-intermediate scatter). This window selects 76,437 events — 22.9% of the uncut sample — by retaining only low-to-moderate momentum transfer, where angular

uncertainty is smallest.

To assess whether the 20-epoch result generalised to the standard 60-epoch convergence used throughout this study, the optimal window was reconstructed to full convergence. The 20-epoch trial of this window during optimisation had produced an apparent $m = 0.477$, nominally below the resolution threshold, but at full 60-epoch convergence the merge metric rose to $m = 0.529$. This illustrates that LM-OSEM over-iteration amplifies noise when event statistics are low, and that optimisation scores obtained at truncated epoch counts must be validated at full convergence before being interpreted as resolution claims. The validated merge metric at 60 epochs ($m = 0.529$) is compared with $m = 0.824$ for the uncut reconstruction at the same convergence (Figure 7). This represents a 35.8% reduction in merge metric. However, the resolved threshold ($m < 0.5$) is not crossed: the 30 mm separation remains unresolved even with optimal event selection.

The improvement is physically consistent. The optimal window excludes events with $E_C > 122.6$ keV, where the Compton angular uncertainty $\delta\theta_c = m_e c^2/[\sin\theta_c\,(E_\gamma - E_C)^2]\,\delta E_C$ grows steeply as $(E_\gamma - E_C) \to 0$. Accepting only $\theta_c \leq 57°$ retains the events with the most reliable cone angles, improving effective angular resolution at the cost of statistics. The fact that the threshold is not crossed indicates that the 30 mm separation lies below the fundamental resolution floor of this geometry: no two-parameter energy window can compensate for the intrinsic PSF width at $r = 300$ mm source-detector distance.

2.5.2 Computational Acceleration

Several implementation choices reduce wall-clock time.

GEANT4 multi-threading. Simulations run using `G4MTRunManager` across 8–32 CPU threads, reducing simulation time approximately proportionally.

GPU-accelerated reconstruction. LM-OSEM is implemented in PyTorch and distributed across multiple GPUs. Each mini-batch is partitioned evenly across available devices; after each batch, per-GPU activity updates are summed. Up to 8 GPUs are used.

Bilinear interpolation. System-matrix lookups use bilinear interpolation in $(E_C, \cos\theta)$ rather than nearest-neighbour, reducing discretisation artefacts without requiring a finer matrix grid.

Adaptive sub-batching. When batches exceed GPU memory, the implementation auto-

matically subdivides and accumulates the ML-EM update over sub-batches before applying the weight update.

3 Conclusions

The simulation and reconstruction framework achieved the primary objectives set out in Section 1.4 and clarified the principal performance limits of the current detector geometry. Four conclusions follow directly from the results.

The detector's intrinsic spatial resolution at 440 keV is approximately 30–35 mm. Two-point sources separated by ≥ 40 mm are cleanly resolved ($m = 0.211$ at 40 mm, $m = 0.029$ at 50 mm); sources at 35 mm are just above the resolution threshold ($m = 0.512$, unresolved); sources at 30 mm are fully merged ($m = 0.824$). Increasing event counts from 10^7 to 2.5×10^8 per source at 30 mm made the merge *more* pronounced, confirming the limit is a fundamental system property rather than a statistical artefact. The observed limit agrees with a first-principles estimate from propagated Compton angular uncertainty combined with depth-of-interaction broadening.

Voxel size must be matched to the system PSF. Reducing voxels from 5 mm to 1 mm on the same 31 mm volume produced diffuse, salt-and-pepper reconstructions, not improved resolution. The cause is system-matrix rank deficiency: sub-PSF voxels are indistinguishable under the detector response, so LM-OSEM distributes activity arbitrarily among degenerate voxels. This is an algorithmic failure mode, not a statistics problem — no achievable event count recovers it.

The three-part activity-concentration metric reliably classifies reconstruction quality. The combined criterion ($\rho \geq 0.5$, $C \geq 200\times$, $\phi \leq 0.20\%$) correctly classified all nine validation cases spanning two voxel sizes and multiple separations. The contrast and hot-background metrics remain valid in the high-coverage regime where the ratio metric alone is unreliable.

Bayesian optimisation of the scatter-energy window reduces the merge metric but does not cross the resolution threshold. The optimal window $[0, 122.6]$ keV ($\theta_c \in [0°, 57°]$, forward-to-intermediate scatter) reduced m from 0.824 to 0.529 at 30 mm separation — a 35.8% improvement confirmed at full 60-epoch convergence. The resolved threshold ($m < 0.5$) was not crossed. As discussed in Section 2.5.1, the 20-epoch trial scores used during optimisation can be optimistic relative to fully converged reconstruc-

tions and must be validated before being interpreted as resolution claims. The optimisation correctly explored the full parameter space and identified the optimal window within 30 trials, with no improvement seen after trial 15.

Clinical context of the 30 mm resolution limit. The intrinsic 30 mm resolution is insufficient for resolving sub-centimetre metastases or performing intra-tumoral dosimetry, but is adequate for two clinically relevant tasks in ^{225}Ac TAT: verifying bulk uptake in centimetre-scale primary lesions, and assessing organ-level biodistribution to detect off-target accumulation in healthy tissue. Progressing to the 5–10 mm resolution required for lesion-level dosimetry would require reducing the inner detector radius or implementing depth-of-interaction measurement to suppress the dominant contribution to angular uncertainty. The first-principles scaling $\delta r \approx r\,\delta\theta_c$ confirmed in Section 2.3.1 provides a quantitative guide for future detector design iterations.

4 Recommendations

1. Perform a noise-free baseline study to isolate resolution contributions. The observed ~ 30 mm resolution limit arises from a combination of Compton angular uncertainty, finite energy resolution ($\sigma_E/E = 3.1\%$), position resolution ($\sigma_{\rm pos} = 5$ mm), and depth-of-interaction uncertainty. Constructing a system matrix with detector smearing disabled and repeating the two-point resolution study will benchmark each contribution separately, isolating the algorithmic floor from the detector-physics floor.

2. Conduct a symmetry and degeneracy analysis. The cylindrical detector geometry has exact azimuthal symmetry: point sources at equal radial distances but different azimuthal angles produce nearly identical event signatures under the position-invariant system-matrix approximation. Simulating point sources at symmetric positions around the detector axis will determine whether this symmetry introduces genuine degeneracy in the reconstruction and how it affects localisation accuracy in the transverse plane.

3. Extend optimisation to the full event-selection parameter space. The current Bayesian optimisation treats only the scatter-energy window. The scatter-to-absorption distance cut, absorption cluster radius, and minimum interaction count are additional candidates where data-driven tuning is likely to yield further resolution improvements.

4. Investigate detector hardware changes to improve angular resolution. A significant component of the ~ 30 mm limit arises from depth-of-interaction uncertainty within

the 0.60 X_0 inner scatter layer. If the physical detector can measure depth of interaction through pulse-shape analysis or segmented readout, incorporating it into the system matrix would directly improve achievable resolution. Reducing the inner-layer thickness is a complementary design change that would lower depth-of-interaction broadening at the cost of scatter efficiency. These hardware-level modifications are the primary levers for reaching the 5–10 mm resolution required for lesion-level dosimetry.

5. Activate the water-box phantom for attenuation-aware studies. All current results assume a vacuum geometry. Patient-representative imaging requires tissue attenuation, which the framework already supports but was not activated in the present study to isolate baseline algorithmic performance in an attenuation-free geometry. A comparison with and without the phantom will quantify the impact of soft-tissue scatter on clinical performance.

6. Investigate deep learning for system-matrix approximation. The KDE-based system matrix is fixed to a single detector configuration, and the position-invariant approximation used in Sec 2.1.1 breaks down near the edges of the field of view. A neural network trained to predict $P(E_C, \cos\theta \mid \mathbf{r})$ directly from source position could provide a continuously differentiable, position-dependent system matrix that accounts for field-of-view variations and reduces the simulation burden required when the detector configuration changes.

5 Deliverables

1. **GEANT4 simulation code (TIIGR)** — C++ source, build system, and macro files for all source and detector configurations used in this work. Delivered via the TIIGR Git repository.

2. **Reconstruction pipeline** — PyTorch-based LM-OSEM implementation with multi-GPU support, HDF5 I/O, YAML-driven configuration, and integrated event-selection pipeline.

3. **System matrices** — Pre-computed KDE and histogram-based system matrices at 440 keV and 218 keV, stored as HDF5 files with full provenance metadata.

4. **Evaluation metric library** — Python implementations of the merge metric, activity-concentration metric, and FWHM evaluation, with validation test cases.

5. **Bayesian optimisation framework** — `ax-platform`-based harness for scatter-energy window optimisation, with crash-recovery and resumption support.

6. **Documentation** — This report, README files for each code module, and a user guide for running end-to-end reconstructions from raw GEANT4 output to analysed images.

7. **Representative output data** — ROOT files for the two-point source resolution study, reconstruction HDF5 epoch files for selected configurations, and plotting scripts used to generate all figures in this report.

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A Glossary

Bayesian optimisation (BO) Sample-efficient method for optimising an expensive black-box function using a probabilistic surrogate model (Gaussian process) updated after each evaluation. The next trial is selected by maximising an acquisition function that balances exploitation and exploration.

Compton camera Radiation imaging device that replaces mechanical collimation with electronic collimation based on Compton scattering kinematics. Each photon interaction constrains the source origin to a cone surface; combining many cones reconstructs the 3D source distribution.

Compton scattering Inelastic scattering of a photon by a loosely bound electron, transferring part of the photon energy to the electron. The scattering angle θ_c is determined by the deposited energy via $\cos\theta_c = 1 - m_e c^2 E_C/[E_\gamma(E_\gamma - E_C)]$.

Detection efficiency (ε) Fraction of simulated primary photons that survive all five event-selection cuts and contribute to reconstruction. For this system at 440 keV, $\varepsilon \approx 1.67\%$.

Expected Improvement (EI) Bayesian optimisation acquisition function that quantifies the expected gain over the current best observation, giving high values where the posterior mean predicts improvement or where posterior uncertainty is large.

Field of view (FOV) The spatial region within which sources can be reliably imaged under the position-invariant system-matrix approximation. Here defined by the 30 mm-radius sphere used for volume-source construction.

FWHM (full width at half maximum) Standard spatial resolution metric: the width of a reconstructed point-source profile at half its peak value. Smaller FWHM indicates better resolution. Computed independently along each Cartesian axis and tracked per epoch.

GEANT4 Open-source Monte Carlo toolkit for particle transport through matter, maintained by CERN. Used here to simulate gamma-ray interactions in the liquid-xenon detector and record per-event interaction data.

Gaussian process (GP) A probabilistic model over functions, fully specified by a mean function and a covariance kernel. The posterior GP conditioned on observations

gives a predictive distribution over unobserved function values, used as the surrogate in Bayesian optimisation.

GPU (graphics processing unit) Massively parallel processor used to accelerate LM-OSEM reconstruction. The implementation distributes mini-batches across up to 8 GPUs, summing per-device activity updates after each batch.

HDF5 Hierarchical Data Format version 5. Portable self-describing binary file format used for system matrices, processed data samples, and per-epoch reconstruction output, each stored with full provenance metadata.

KDE (kernel density estimation) Non-parametric method for estimating an unknown probability density from finite samples by summing smooth kernels centred at each data point. Used here in 2D over $(E_C, \cos\theta)$ with an Epanechnikov kernel to build the system matrix.

LM-OSEM List-mode ordered-subsets expectation maximisation. Iterative reconstruction algorithm that processes events individually and updates the activity estimate using subsets of events per iteration. Ordered subsets accelerate convergence relative to full-dataset EM.

Merge metric (m) Continuous valley-to-peak ratio on the marginal activity profile, ranging from 0 (fully resolved) to 1 (fully merged). Sources are declared resolved when $m < 0.5$, consistent with the Rayleigh criterion.

PSF (point spread function) The reconstructed image of an idealised point source; characterises the spatial blurring introduced by the combined detector and algorithm response. The intrinsic PSF of this system has ~ 30 mm FWHM at 440 keV.

Rayleigh criterion Classical resolution criterion: two point sources are just resolved when the central maximum of one coincides with the first minimum of the other. Approximated here by the threshold $m < 0.5$ on the merge metric.

ROOT Object-oriented data analysis framework from CERN. Used for storing raw GEANT4 simulation output as `TTree` objects containing per-event and per-interaction data.

SPECT (single-photon emission computed tomography) Nuclear medicine imaging modality that reconstructs the 3D distribution of gamma-emitting radiotracers using mechanically collimated detectors. The mechanical collimator rejects most incident photons, producing an inherent sensitivity–resolution trade-off.

System matrix (A) Forward model relating voxel source activity to detector measurement probabilities. Element $A_j(y_n)$ gives the probability that detected event y_n originated from voxel j. Built from a volume-source simulation by KDE over $(E_C, \cos\theta)$ under the position-invariant approximation.

TAT (targeted alpha therapy) Radiopharmaceutical cancer treatment linking an alpha-emitting radionuclide to a tumour-targeting biological carrier. The alpha particles deliver highly localised radiation with short range in tissue. ^{225}Ac is the primary radionuclide of interest in this work.

TIIGR Therapeutic Isotope Imager with Gamma Rays. The TRIUMF Compton-camera system for which this project's simulation and reconstruction framework was developed.

TRIUMF Canada's national particle accelerator centre, located on the UBC campus in Vancouver, British Columbia. Project sponsor.

Voxel Volumetric pixel; the smallest discrete unit of the 3D reconstruction grid. Voxel size must be matched to the system PSF: sub-PSF voxels produce rank-deficient system matrices and noise-dominated reconstructions.

B GEANT4 Simulation Setup

B.1 Primary Generator

The primary generator emits one gamma-ray photon per event via `G4ParticleGun`. The angular distribution is fully isotropic over 4π sr, sampled by $\cos\theta$ uniformly on $[-1, 1]$ and ϕ uniformly on $[0, 2\pi)$. Source energies of 440 keV and 218 keV correspond to gamma lines in the ^{225}Ac decay chain. The framework supports point sources, ring sources, spherical volume sources, and multi-point configurations.

B.2 Detector Geometry

The world volume is a $3\times3\times3$ m^3 air box centred at the origin. The detector consists of two concentric cylindrical shells (`G4Tubs`) coaxial with the z-axis, each with half-length 75 cm. The inner shell has inner radius 30 cm and thickness 0.60 X_0 in liquid xenon; the outer

shell has inner radius 50 cm and thickness 1.60 X_0 in liquid xenon. Each sensitive volume is enclosed in a 3 mm aluminium structural shell, and the inter-shell gap is filled with liquid nitrogen. An optional water-box phantom (simplified human body) is implemented in the framework but was not activated here.

B.3 Physics List

The physics list registers three modules: `G4EmLowEPPhysics` (low-energy EM: Compton, photoelectric, pair production, Rayleigh), `G4DecayPhysics` (particle decays), and `G4RadioactiveDecayPhysics` (nuclear radioactive decay). The production cut is 0.1 mm for all particles in all regions.

B.4 Stepping Action and Recorded Quantities

Only the primary gamma (track ID = 1) is followed. At each step with positive energy deposition $\Delta E = E_{\mathrm{pre}} - E_{\mathrm{post}} > 0$, the following quantities are recorded: interaction position, energy deposited, scattering angle, global time, interaction type (Compton, photoelectric, or other), cumulative interaction index, and detector layer. Rayleigh steps are flagged but carry no energy deposition.

B.5 Event Selection and ROOT Output

An optional event-selection mode applies topological cuts at simulation time. Secondary particles are killed immediately, and only events matching the canonical Compton scatter $\rightarrow$ photoelectric absorption topology are retained (scatter-to-absorption separation > 100 mm, absorption cluster radius ≤ 3 mm). Two `TTree` objects are written per run: a primary tree storing per-event and per-interaction data, and a metadata tree storing run-level parameters.

B.6 Simulation Execution

The TIIGR executable accepts a macro file, random seed, and thread count for multi-threaded execution via `G4MTRunManager`. A schematic macro structure is shown below.

```
# --- Inner detector layer ---
/TIIGR/det/configureNewDetector
/TIIGR/det/material    lXe
/TIIGR/det/innerRadius 30 cm
/TIIGR/det/thicknessX0 0.60
/TIIGR/det/shellThickness 3 mm
/TIIGR/det/nitrogenGap
/TIIGR/det/addDetector

# --- Outer detector layer ---
/TIIGR/det/configureNewDetector
/TIIGR/det/material    lXe
/TIIGR/det/innerRadius 50 cm
/TIIGR/det/thicknessX0 1.60
/TIIGR/det/shellThickness 3 mm
/TIIGR/det/addDetector

/output/fileName <run_name>
/run/initialize
/TIIGR/source/energy   440 keV
/TIIGR/source/position <x> <y> <z> mm
/run/beamOn <nEvents>
```

C Data Smearing

To make the simulated GEANT4 data more representative of measurements from a physical detector, the idealised interaction data were degraded using a smearing procedure. The raw simulation outputs provide truth information: interaction positions and deposited energies are recorded without the uncertainty that would be present in an experimental system. In practice, detector readout is subject to finite spatial and energy resolution. Smearing was therefore applied to approximate these effects and produce data that better reflect realistic detector performance.

C.1 Energy Smearing

Energy smearing was introduced by modelling the measured deposited energy as a random variable drawn from a normal distribution centred at the true deposited energy. The width of this distribution was scaled from a reference full width at half maximum (FWHM) specified at 662 keV. For a deposited energy E_{measured}, the corresponding energy broadening was taken as

$$\Delta E_{\text{measured}} = \frac{\text{FWHM}}{100}\sqrt{662}\sqrt{E_{\text{measured}}}, \tag{1}$$

where the FWHM value is given as a percentage at 662 keV. The smeared energy was then sampled from a Gaussian distribution with mean E_{measured} and standard deviation $\Delta E_{\text{measured}}/(2\sqrt{2\ln 2})$. This approach captures the expected increase in absolute uncertainty with increasing deposited energy.

C.2 Position Smearing

The simulated interaction positions were originally produced in Cartesian coordinates. To model finite spatial resolution, each coordinate was perturbed independently by adding Gaussian noise with zero mean. The standard deviation of this noise was related to the detector spatial resolution R through

$$\sigma = \frac{R}{\sqrt{3}}. \tag{2}$$

This was done separately for each spatial coordinate so that the interaction location was displaced randomly in three dimensions. Boundary checks were applied to ensure that smeared positions remained within the physical detector volume.

C.3 Resolution Values Used

For both argon and xenon, a spatial resolution of 5 mm was assumed. The energy resolution at 662 keV was taken to be 4.1% for argon and 3.1% for xenon. These values were used to define the Gaussian broadening applied during the smearing stage.

D Bayesian Optimisation: Mathematical Detail

D.1 Gaussian Process Surrogate

Let $\mathbf{x} = (E_{C,\min}, E_{C,\max})$ and $f(\mathbf{x}) = m(\mathbf{x})$. BO models f as a Gaussian process [17]: $f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'; \boldsymbol{\theta}))$. The kernel is Matérn-5/2 [19], encoding that f is twice mean-square differentiable:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{\ell} + \frac{5r^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}r}{\ell}\right), \quad r = \|\mathbf{x} - \mathbf{x}'\|.$$

After t evaluations $\mathcal{D}_t = \{(\mathbf{x}_i, y_i)\}_{i=1}^t$, the posterior predictive mean and variance at $\mathbf{x}_*$ are

$$\begin{aligned}
\mu_t(\mathbf{x}_*) &= \mu(\mathbf{x}_*) + \mathbf{k}_*^{\mathrm{T}}(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}(\mathbf{y} - \boldsymbol{\mu}), \\
\sigma_t^2(\mathbf{x}_*) &= k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^{\mathrm{T}}(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}\mathbf{k}_*.
\end{aligned}$$

Observation noise $\sigma_n^2 = 0$ (deterministic objective).

D.2 Hyperparameter Estimation and Acquisition

Kernel hyperparameters $\boldsymbol{\theta} = \{\ell, \sigma_f^2\}$ are inferred by maximising the log marginal likelihood:

$$\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = -\tfrac{1}{2}\mathbf{y}^{\mathrm{T}}\mathbf{K}^{-1}\mathbf{y} - \tfrac{1}{2}\log|\mathbf{K}| - \tfrac{t}{2}\log 2\pi.$$

The log-determinant term provides implicit regularisation, favouring smoother explanations unless data strongly support finer structure.

The Expected Improvement acquisition function [6] is

$$\mathrm{EI}(\mathbf{x}) = (f^* - \mu_t(\mathbf{x}))\Phi(Z) + \sigma_t(\mathbf{x})\phi(Z), \quad Z = \frac{f^* - \mu_t(\mathbf{x})}{\sigma_t(\mathbf{x})},$$

where $f^* = \min_{i \le t} y_i$ and Φ, ϕ are the standard normal CDF and PDF. EI balances exploitation (high when the posterior mean predicts improvement) against exploration (high when posterior variance is large).

D.3 Initialisation

The first $n_{\text{init}} = 6$ trials use a quasi-random sequence, which provides more uniform parameter-space coverage than uniform random sampling and avoids early trapping in local regions.

E Event-Count Estimation for Point Sources

The number of GEANT4 events simulated for point-source studies is motivated by the expected number of gamma photons emitted from the tumour during acquisition. The total number of simulated events is

$$N_{\text{events}} = C\,V\,T_{\text{scan}}\,\text{BR},$$

where C is the ^{225}Ac activity concentration (Bq/mL), V is the tumour volume (mL), T_{scan} is scan duration (s), and BR is the branching ratio for the gamma line being simulated. Activity concentration is determined from the injected dose, tumour uptake fraction, and decay at the time of imaging:

$$A(t) = A_0 \exp\left(-\ln 2 \cdot \frac{t}{T_{1/2}}\right),$$

with $T_{1/2} = 9.9$ days for ^{225}Ac. Modelling the source as a sphere of radius r gives $V = \frac{4}{3}\pi r^3$; at soft-tissue density ~ 1 g/cm^3, volume in mL is numerically equal to mass in grams.

For the 440 keV gamma ray emitted by ^{213}Bi (a daughter of ^{225}Ac), BR ≈ 0.261 [10]. Taking $C = 100$ kBq/mL, $V = 100$ mL ($r \approx 29$ mm), $T_{\text{scan}} = 600$ s:

$$N_{\text{events}} \approx 1.57 \times 10^9.$$

For a smaller 10 mL tumour ($r \approx 13$ mm), $N_{\text{events}} \approx 1.57 \times 10^8$. These estimates explain why simulation counts of 5×10^7 to 5×10^8 events are physically reasonable for this study.

F Quality Metric: Threshold Calibration

Integration region half-width. The value $h = 10$ mm is chosen empirically: well-converged 5 mm reconstructions consistently concentrate $\sim 88\%$ of total activity within this region, while the expected ratio for a uniform-noise image equals $f_{\mathrm{mask}} \approx 0.2\%$ — a $\sim 400\times$ margin. Note that $h = 10$ mm does not correspond to a theoretical PSF width: the raw Compton PSF is ~ 30 mm FWHM [21], but LM-OSEM concentrates activity far more sharply than the raw PSF suggests due to the resolution-recovery property of ML reconstruction [20, 8].

Hot-background threshold selection. The threshold $\alpha = 0.05$ classifies a background voxel as hot if activity exceeds 5% of the mean per mask voxel assuming all image energy were concentrated in $\mathcal{C}$. The reference quantity $\Lambda_{\mathrm{total}}/|\mathcal{C}|$ uses total image energy rather than local sum, decoupling the threshold from reconstruction quality: if activity leaks out of the mask, a locally-referenced threshold would drop and mask the very noise we aim to detect [18].

Adaptive ratio threshold. When mask fraction f_{mask} is large (tight separations on small grids), the fixed threshold $\rho_{\mathrm{thresh}} = 0.5$ may be exceeded by noisy images since $\rho \approx f_{\mathrm{mask}}$ under uniform noise [14]. The adaptive threshold

$$\rho_{\mathrm{thresh}}^{\mathrm{adapt}} = \max(0.5,\ 3f_{\mathrm{mask}})$$

maintains at least a $3\times$ margin above the expected noise baseline. A `high_coverage` flag is raised whenever $f_{\mathrm{mask}} > 20\%$.

Validation. Empirical calibration against nine reconstruction configurations (two voxel sizes, multiple source separations, two epoch counts) correctly classified all nine cases. The 1 mm voxel reconstructions at epoch 60 — with $\rho \approx 0.74$–0.81 comparable to the 5 mm cases and declared GOOD by ρ alone — are correctly identified as BAD by contrast ($C \approx 80$–$120 \ll 200$) and hot-background fraction ($\phi \approx 0.25$–$0.36\% > 0.20\%$). Thresholds were calibrated for this specific detector geometry and may require retuning for substantially different configurations [2].